# ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** cell | Cell. |
| 2 | **(c)** water | Through a membrane. |
| 3 | **(b)** two identical cells | Growth. |
| 4 | **(b)** Robert Hooke | 1665. |
| 5 | **(a)** high to low concentration | Gradient. |
| 6 | **(c)** gametes | Half chromosomes. |
| 7 | **(d)** nucleus | Contains DNA. |
| 8 | **(b)** swells | Water enters. |
| 9 | **(c)** 46 | 23 pairs. |
| 10 | **(c)** cell wall | Cellulose. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Basic structural and functional unit of life (2). | 2 |
| 12 | Cells without a nucleus (1); Bacteria (1). | 1+1 |
| 13 | Photosynthesis (2). | 2 |
| 14 | Schleiden (1); Schwann (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Cell wall (1); Plastids (1); Vacuole size (1). | 1+1+1 |
| 16 | Protects (1); Controls entry (1); Controls exit (1). | 1+1+1 |
| 17 | Nucleus (1); Mitochondria (1); Chloroplasts (1). | 1+1+1 |
| | **OR:** Nucleus (1); Mitochondria (1); Ribosomes (1). | 1+1+1 |
| 18 | Control centre (1); Contains DNA (1); Heredity (1). | 1+1+1 |
| | **OR:** Made of cells (1); Basic unit (1); From pre-existing cells (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Membrane (1); Nucleus (1); Cytoplasm (1); Organelles (1); Functions (1). | 1+1+1+1+1 |
| | **OR:** Process (1); Two cells (1); Growth (1); Repair (1); Same chromosomes (1). | 1+1+1+1+1 |
| 20 | Wall (1); Membrane (1); Plastids (1); Vacuole (1); Nucleus (1). | 1+1+1+1+1 |
| | **OR:** Process (1); Four cells (1); Half chromosomes (1); Gametes (1); Variation (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** shrinks | Water leaves. |
| 2 | **(b)** ribosomes | Ribosome subunits. |
| 3 | **(c)** mitochondria | Energy. |
| 4 | **(b)** swell | Endosmosis. |
| 5 | **(d)** DNA | Heredity. |
| 6 | **(b)** ribosomes | Synthesis. |
| 7 | **(b)** shrinking of cytoplasm | In hypertonic solution. |
| 8 | **(b)** plant cells | Plants. |
| 9 | **(d)** lysosomes | Digestive. |
| 10 | **(c)** prokaryotes | No nucleus. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Cells with a true nucleus (1); Plant cells (1). | 1+1 |
| 12 | Lysosome (2). | 2 |
| 13 | Movement of water through a selectively permeable membrane (2). | 2 |
| 14 | Gives shape and protection (2). | 2 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Chloroplast (1); Chromoplast (1); Leucoplast (1). | 1+1+1 |
| 16 | Storage (1); Turgidity (1); Waste storage (1). | 1+1+1 |
| 17 | Transport (1); Protein synthesis (rough) (1); Lipid synthesis (smooth) (1). | 1+1+1 |
|  | **OR:** Amoeba (1); Paramecium (1); Bacteria (1). | 1+1+1 |
| 18 | Number of cells (1); Chromosome number (1); Purpose (1). | 1+1+1 |
|  | **OR:** Function (1); Examples (1); Structure fits function (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Diffusion (1); Osmosis (1); Examples (1); Differences (1); Importance (1). | 1+1+1+1+1 |
|  | **OR:** Hollow potato (1); Sugar solution (1); Water (1); Level rises (1); Osmosis (1). | 1+1+1+1+1 |
| 20 | Statement (1); Schleiden (1); Schwann (1); Virchow (1); Importance (1). | 1+1+1+1+1 |
|  | **OR:** Raisins in water (1); Raisins in sugar solution (1); Swells and shrinks (1); Osmosis (1); Conclusion (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** colour | Pigments. |
| 2 | **(c)** chloroplasts | Chlorophyll. |
| 3 | **(b)** cell walls | Cellulose. |
| 4 | **(c)** starch | Colourless. |
| 5 | **(d)** Amoeba | One cell. |
| 6 | **(a)** plant cells | Storage. |
| 7 | **(d)** packages materials | Dispatch. |
| 8 | **(a)** selectively permeable | Controls passage. |
| 9 | **(a)** nucleus | DNA. |
| 10 | **(c)** a network of membranes | Transport. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Division giving two identical daughter cells (2). | 2 |
| 12 | Movement of particles from higher to lower concentration (2). | 2 |
| 13 | Protein synthesis (2). | 2 |
| 14 | Division giving gametes with half the chromosomes (2). | 2 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Raisins in water (1); Roots absorbing water (1); Wilting (1). | 1+1+1 |
| 16 | Swells (1); May burst (1); Haemolysis (1). | 1+1+1 |
| 17 | Nuclear membrane (1); Nucleolus (1); Chromatin (1). | 1+1+1 |
| | **OR:** Smell of food (1); Ink in water (1); Gas exchange (1). | 1+1+1 |
| 18 | Water leaves (1); Cytoplasm shrinks (1); Plasmolysis (1). | 1+1+1 |
| | **OR:** Everything made of cells (1); Cells perform functions (1); Cells divide (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Nucleus (1); Size (1); Organelles (1); Examples (1); Complexity (1). | 1+1+1+1+1 |
| | **OR:** Hypertonic solution (1); Water leaves (1); Cytoplasm shrinks (1); Wall remains (1); Example (1). | 1+1+1+1+1 |
| 20 | Nucleus (1); Mitochondria (1); Ribosomes (1); Golgi (1); Lysosomes (1). | 1+1+1+1+1 |
| | **OR:** Shape (1); Protection (1); Prevents bursting (1); Only in plants (1); Cellulose (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.